Hugo Marquerie 05/01/2026

La preimagen de una subvariedad por un morfismo es una subvariedad

Proposición 1. *Sea $f\colon X \to Y$ un morfismo de v.a.a. y $W \subset Y$ una subvariedad*

$$\implies \exists! J \subset K[x_1, \dots, x_n] \text{ ideal} : \mathbb{I}(X) \subset J \wedge \iota_1^*(J) = \langle (f^* \circ \iota_2^*)(\mathbb{I}(W)) \rangle \wedge f^{-1}(W) = \mathbb{V}(J).$$

En particular, $f^{-1}(W)$ es una subvariedad algebraica afín de X.

***Demostración*:** Sea $a = (a_1, \dots, a_n) \in X$. Entonces, $f(a) \in W$

$$\iff \forall G \in \mathbb{I}(W) : G(f(a)) = G(f_1(\overline{x_1}, \dots, \overline{x}_n), \dots, f_m(\overline{x}_1, \dots, \overline{x}_n))(a_1, \dots, a_n) = 0$$
$$= (f^* \circ \iota_2^*)(G)(a_1, \dots, a_n) = 0.$$

Esto es equivalente a que $\forall H \in (f^* \circ \iota_2^*)(\mathbb{I}(W)) : H(a) = 0$.

Si consideramos el ideal $\langle (f^* \circ \iota_2^*)(\mathbb{I}(W)) \rangle \subset K[X]$, tenemos que solo existe un único ideal $J \subset K[x_1, \dots, x_n]$ que contiene a $\mathbb{I}(X)$ y tal que $\iota_1^*(J) = \langle (f^* \circ \iota_2^*)(\mathbb{I}(W)) \rangle$, que es

$$J = (\iota_1^*)^{-1} \left(\langle (f^* \circ \iota_2^*)(\mathbb{I}(W)) \rangle \right).$$

Por tanto, $f^{-1}(W) = \mathbb{V}(J)$ y $f^{-1}(W)$ es una subvariedad algebraica afín de X. ■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.